

IT Portfolio – Steven Nielsen

This portfolio showcases hands-on technical labs demonstrating skills in Windows Server, Active Directory, Networking, Virtualization, Cloud Identity, Ticketing, VPN, and SaaS troubleshooting. Each lab includes objectives, tooling, and a summary of the work performed. Screenshots are available in the associated GitHub repository.

Labs Included:

- Lab 1 – Windows Server 2019 + Active Directory Domain Deployment
- Lab 2 – DNS, DHCP, and Group Policy Management
- Lab 3 – IT Ticketing & Troubleshooting Workflow Simulation
- Lab 4 – VPN & Network Connectivity Troubleshooting Lab
- Lab 5 – SaaS / Cloud Identity & MFA Support Lab

Lab 1 – Windows Server 2019 + Active Directory Domain Deployment

Objective: Deploy a Windows Server 2019 virtual machine, promote it to an Active Directory Domain Controller, and configure a basic domain environment with organizational units and user accounts.

Tools: VirtualBox, Windows Server 2019 (evaluation ISO), Windows 10 ISO.

Summary of Work Performed:

- Created a new VirtualBox VM (DC01) with 8 GB RAM and 60 GB storage, attached the Windows Server 2019 ISO, and completed installation.
- Configured a static IP address on the server and set DNS to the local host address in preparation for domain services.
- Installed the Active Directory Domain Services and DNS Server roles, then promoted the server to a new forest with the domain name lab.local.
- Created organizational units (IT, HR, Finance, StandardUsers) in Active Directory Users and Computers.
- Created multiple test user accounts and a delegated admin account, and verified successful authentication from a Windows 10 domain-joined client.
- Documented each configuration step and captured screenshots to demonstrate a working domain environment.

Lab 2 – DNS, DHCP, and Group Policy Management

Objective: Configure DNS and DHCP services in a Windows Server environment and apply Group Policy Objects (GPOs) to manage client settings.

Tools: Windows Server 2019, Windows 10 domain client, VirtualBox networking.

Summary of Work Performed:

- Configured a DHCP scope on the domain controller to hand out IP addresses, default gateway, and DNS settings to domain-joined clients.
- Validated dynamic IP assignment from the DHCP server on a Windows 10 client and confirmed name resolution via DNS.
- Created and linked a Group Policy Object to enforce password complexity, screen lock timeout, and desktop restrictions for StandardUsers.
- Forced a Group Policy update on client systems and documented resulting changes.
- Simulated and resolved common DNS and DHCP issues (incorrect DNS server, duplicate IP, stale records) and documented troubleshooting steps.

Lab 3 – IT Ticketing & Troubleshooting Workflow Simulation

Objective: Demonstrate structured troubleshooting and documentation skills using a ticketing system to track user issues from intake to resolution.

Tools: Jira Service Management or Freshdesk (free tier), Windows 10 test client, browser-based SaaS tools.

Summary of Work Performed:

- Configured a small help desk project with request types for login issues, VPN connectivity, email problems, and performance complaints.
- Created sample tickets to represent real-world issues such as 'User cannot log in – account locked,' 'VPN disconnects intermittently,' and 'Computer is slow.'
- Documented each troubleshooting session using clear, step-by-step updates, including root cause, actions taken, and final resolution.
- Applied consistent categorization, priority, and SLA-style time tracking to demonstrate familiarity with ticket triage best practices.
- Exported ticket views and resolution notes as evidence of strong written communication and documentation skills.

Lab 4 – VPN & Network Connectivity Troubleshooting Lab

Objective: Configure a simple VPN connection and practice diagnosing common network connectivity problems affecting remote users.

Tools: OpenVPN or WireGuard, Windows Server 2019, Windows 10 client, VirtualBox host-only/bridged networking.

Summary of Work Performed:

- Deployed a lightweight VPN server on the lab network and generated a client profile.
- Installed the VPN client on a Windows 10 VM and verified successful connection to internal resources via ping and file share access.
- Simulated connectivity problems by blocking ports, misconfiguring DNS, and disabling routes, then observed impact on name resolution and reachability.
- Performed step-by-step troubleshooting by checking IP configuration, routing tables, DNS settings, and firewall rules.
- Documented each failure scenario and the corrective action taken, emphasizing a calm and methodical approach to network issues.

Lab 5 – SaaS / Cloud Identity & MFA Support Lab

Objective: Practice cloud identity administration and SaaS troubleshooting with a focus on user access, MFA policies, and login support.

Tools: Microsoft 365 Developer tenant (Entra ID / Azure AD), browser, sample test accounts.

Summary of Work Performed:

- Created a Microsoft 365 developer tenant and provisioned multiple test users with different roles and licenses.
- Enabled Multi-Factor Authentication (MFA) and Conditional Access rules to require additional verification for selected users.
- Simulated common user issues such as lost MFA device, incorrect passwords, and disabled accounts, then resolved them via the admin portal.
- Tracked changes using the audit logs and sign-in logs to correlate user reports with backend events.
- Documented standard operating procedures (SOPs) for password resets, MFA resets, and basic SaaS access troubleshooting.

Screenshots & Source Files

Full screenshots, configuration exports, and step-by-step walkthroughs for each lab are available in the accompanying GitHub repository. This PDF is intended as a high-level portfolio overview for recruiters and hiring managers.